

ECC

Alex Chai

Anthony Chang

Erin Jaen

Henry Yuen

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Chapter 1

Linear Codes

1.1 Definitions

Definition 1 (subspace-def). Let q be a prime power (i.e. $q = p^s$ for some prime p and integer $s \geq 1$). $C \subseteq \mathbb{F}_q^n$ is a linear code if it is a linear subspace of F_q^n . If C has dimension k and distance d then it will be referred to as an $[n, k, d]_q$ or just an $[n, k]_q$ code.